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BACHELOR THESIS

By

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Title:

Development of an Internal AI Chatbot for Company Knowledge Retrieval

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To whom it may concern,

I, **Tràn Đình Chí** (supervisor's name), certify that the thesis/ internship report of Mr/Ms. **Nguyen Duy Ngoc** (student's name) is qualified to be presented in the Internship Jury 2025–2026.

Hanoi, __/__/______

Supervisor's signature

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List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
BM25	Best Match 25
CPU	Central Processing Unit
CUDA	Compute Unified Device Architecture
GGUF	GPT-Generated Unified Format
GPU	Graphics Processing Unit
HR	Human Resources
KB	Knowledge Base
LLM	Large Language Model
LoRA	Low-Rank Adaptation
NLP	Natural Language Processing
QLoRA	Quantized Low-Rank Adaptation
RAG	Retrieval-Augmented Generation
RRF	Reciprocal Rank Fusion
SLM	Small Language Model
UI	User Interface
USTH	University of Science and Technology of Hanoi
VRAM	Video Random Access Memory

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Abstract

This thesis presents the design and implementation of a fully local, offline Retrieval-Augmented Generation (RAG) chatbot for Vietnamese company knowledge retrieval. The system enables employees to query internal Human Resources (HR) policies in Vietnamese without relying on cloud-based language model services, thereby preserving data privacy and eliminating per-query API costs.

The knowledge base consists of 20 Vietnamese HR policy documents covering topics such as leave policy, employment contracts, salary, social insurance, and workplace regulations. Eight documents were authored manually and twelve were generated using the Minimax-M2.7 large language model API. Documents are chunked using a recursive character text splitter and embedded with the `paraphrase-multilingual-MiniLM-L12-v2` sentence-transformer model, then stored in a persistent ChromaDB vector database.

A key contribution of this work is a hybrid retrieval strategy that combines BM25 keyword search with vector cosine similarity search, merged via Reciprocal Rank Fusion (RRF). This approach significantly improves retrieval accuracy for short Vietnamese queries, which fail with pure vector search due to low embedding specificity. Answer generation uses the Phi-3-Mini-4k-instruct model in GGUF format, running locally via `llama-cpp-python`.

Additionally, a QLoRA fine-tuning experiment was conducted on 3,343 synthetic question-answer pairs. The experiment produced catastrophic hallucinations due to severely contaminated training data (only 37% domain-relevant content), and is reported as a negative result with root cause analysis.

Experimental results demonstrate that hybrid BM25+vector retrieval with RRF outperforms pure vector search on Vietnamese queries, achieving correct source attribution in cases where the baseline fails.

Keywords: Retrieval-Augmented Generation, Large Language Model, Vietnamese NLP, BM25, vector search, knowledge retrieval.

1. Introduction

1.1 Context and Motivation

The rapid advancement of Large Language Models (LLMs) has created new possibilities for automated knowledge retrieval in enterprise environments [1]. In Vietnamese companies, HR policy documents are typically long, verbose, and available only in PDF or printed format, requiring employees to spend 30–60 minutes searching for a specific policy answer. Cloud-based solutions such as ChatGPT or Gemini could theoretically address this problem, but introduce serious concerns: (1) sensitive HR data must be transmitted to third-party servers, posing a data privacy risk; (2) per-query API costs scale poorly with company-wide usage; and (3) responses are not grounded in the company's specific internal documents.

Retrieval-Augmented Generation (RAG) [1] offers a principled solution: rather than relying solely on the LLM's parametric knowledge, the system first retrieves relevant document passages and injects them into the prompt context. This grounds the generated answer in verifiable source material and enables the use of a smaller, locally-run model, as the retrieval step compensates for the model's limited domain knowledge.

1.2 Problem Statement

This work addresses the following specific challenges:

1. Vietnamese employees cannot quickly access HR policy answers without manually reading through long documents.
2. Cloud LLMs are unsuitable for internal HR queries due to privacy constraints and cost.
3. Standard vector similarity search performs poorly on short Vietnamese queries (3–6 tokens), where keyword overlap is more discriminative than embedding similarity.
4. Fine-tuning a domain-specific model requires curated, high-quality training data, which is difficult to obtain for Vietnamese HR content.

1.3 Literature Review

Retrieval-Augmented Generation. Lewis et al. [1] introduced RAG as a general architecture combining a dense retrieval component with a sequence-to-sequence generator. The retriever fetches relevant passages from an external knowledge corpus, and the generator conditions on both the question and the retrieved passages to produce grounded answers. This architecture has become the dominant approach for knowledge-intensive NLP tasks.

Lexical Retrieval: BM25. BM25 (Best Match 25) [2] is a probabilistic ranking function that scores documents based on term frequency and inverse document frequency, with saturation and length normalization. Despite its simplicity, BM25 remains highly competitive for short, keyword-focused queries and serves as a strong baseline in information retrieval benchmarks.

Hybrid Retrieval and Reciprocal Rank Fusion. Cormack et al. [3] demonstrated that combining rankings from multiple retrieval systems via Reciprocal Rank Fusion (RRF) consistently outperforms individual systems. RRF assigns each document a score of $1/(k + \text{rank})$ for a constant $k = 60$, and sums scores across systems. The approach is robust, parameter-light, and does not require score normalization.

Small Language Models. Microsoft's Phi-3-Mini [4] is a 3.8-billion-parameter model demonstrating strong instruction-following capability, particularly suited for resource-constrained deployment. Its 4-bit GGUF quantization reduces memory requirements to approximately 2.3 GB, enabling CPU-only inference on consumer hardware.

Multilingual Sentence Embeddings. Reimers and Gurevych [5] introduced Sentence-BERT, a modification of the BERT architecture for producing semantically meaningful sentence embeddings via siamese networks. The `paraphrase-multilingual-MiniLM-L12-v2` model extends this approach to 50+ languages, including Vietnamese, with 384-dimensional embeddings.

Parameter-Efficient Fine-Tuning. LoRA [7] enables efficient adaptation of pre-trained LLMs by injecting low-rank decomposition matrices into transformer layers, dramatically reducing trainable parameters. QLoRA [6] extends this to 4-bit quantized models, enabling fine-tuning on consumer GPUs with 4–8 GB VRAM.

Vietnamese NLP. Nguyen et al. [10] highlight key challenges in Vietnamese NLP: the language uses diacritics to distinguish tone and meaning, lacks whitespace-delimited word boundaries in all cases, and has limited pre-trained resource availability compared to English. These challenges motivate the use of multilingual embedding models rather than monolingual ones.

1.4 Contribution of This Work

This thesis makes the following contributions:

1. A fully local, offline Vietnamese HR chatbot using RAG, requiring no internet connectivity at inference time.
2. A 20-document Vietnamese HR knowledge base combining manually authored and LLM-generated policy documents, covering the most common HR topics in Vietnamese companies.
3. A hybrid BM25 + vector retrieval pipeline with RRF fusion, implemented in `src/hybrid_retriever` demonstrating improved retrieval accuracy on short Vietnamese queries.
4. A documented QLoRA fine-tuning negative result, providing a concrete case study of data quality requirements for domain adaptation.

2. Objectives

The scientific objectives of this thesis are:

1. **Design and implement** a fully local RAG chatbot for Vietnamese HR knowledge retrieval, capable of answering employee questions in Vietnamese without any cloud API dependency.

2. **Evaluate** hybrid BM25 + vector retrieval with Reciprocal Rank Fusion against a pure vector search baseline, demonstrating improved accuracy on short Vietnamese queries.

3. **Investigate and document** QLoRA fine-tuning as an alternative approach to RAG for Vietnamese HR domain adaptation, including a root cause analysis of the negative result obtained.

The strategy adopted is to build the system incrementally: first establish the retrieval pipeline and knowledge base, then integrate the local LLM for generation, and finally evaluate both the retrieval component in isolation and the end-to-end system on representative Vietnamese HR queries.

3. Materials and Methods

3.1 System Architecture

The system follows a standard RAG architecture with three main stages: (1) offline knowledge base construction, (2) online hybrid retrieval, and (3) answer generation with a local LLM. Figure 3.1 presents the end-to-end pipeline.

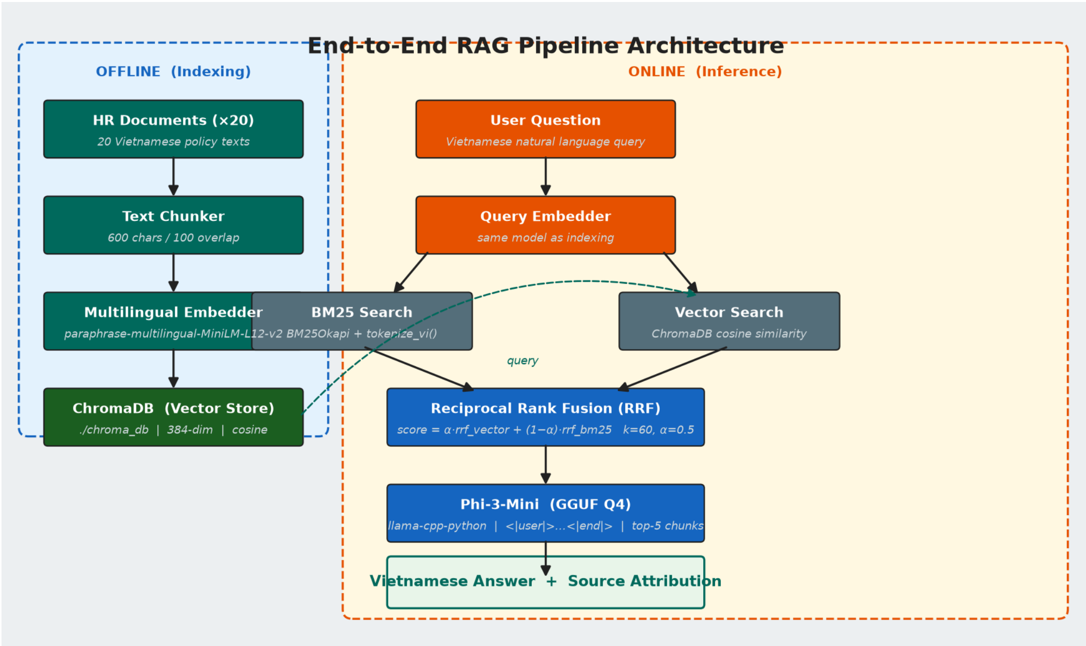

Figure 3.1: *End-to-end system architecture. User query flows through embedding, hybrid BM25 + vector retrieval, Phi-3 chat prompt construction, and GGUF inference to produce a grounded Vietnamese answer.*

At inference time, a user query is embedded into a 384-dimensional vector and used to retrieve the top-k most relevant document chunks from a persistent ChromaDB vector store, augmented by BM25 keyword ranking. The top-5 chunks are injected into a Phi-3-Mini prompt template and the model generates a Vietnamese answer grounded in the retrieved context.

3.2 Knowledge Base Construction

3.2.1 Document Authoring

Eight Vietnamese HR policy documents were authored manually, covering the most frequently queried HR topics in Vietnamese workplaces:

(1) `chinh_sach_nghi_phep.txt` — Annual leave policy

(2) `hop_dong_lao_dong.txt` — Employment contract terms

(3) `chinh_sach_luong.txt` — Salary and payroll policy

(4) `noi_quy_cong_ty.txt` — General company regulations

(5) `ky_luat_lao_dong.txt` — Labor discipline and sanctions

(6) `phuc_loi_nhan_vien.txt` — Employee benefits

(7) `tuyen_dung_dao_tao.txt` — Recruitment and training

(8) `bao_hiem_xa_hoi.txt` — Social insurance contributions

Each document was written with explicit factual content to support direct retrieval. For example, the leave policy explicitly states "1 ngày nghỉ phép mỗi tháng" (1 leave day per month) rather than only "12 ngày mỗi năm" (12 days per year), enabling the model to answer queries about monthly leave without needing to perform implicit arithmetic.

3.2.2 AI-Generated Documents

Twelve additional documents were generated using the Minimax-M2.7 large language model [11] via API. The generation prompt provided a topic description and required the output to conform to Vietnamese HR policy writing conventions. The Minimax-M2.7 model is a reasoning model that outputs `<think>...</think>` blocks before the final content; a post-processing step stripped these reasoning tokens before saving each document.

Topics generated include: overtime policy, maternity and parental leave, remote work policy, business travel guidelines, performance review process, IT security policy, special public holidays, anti-harassment policy, resignation and handover procedures, anti-corruption policy, workplace health and safety, and allowances and subsidies.

The final knowledge base totals 20 Vietnamese HR policy documents, as summarised in Figure 3.2.

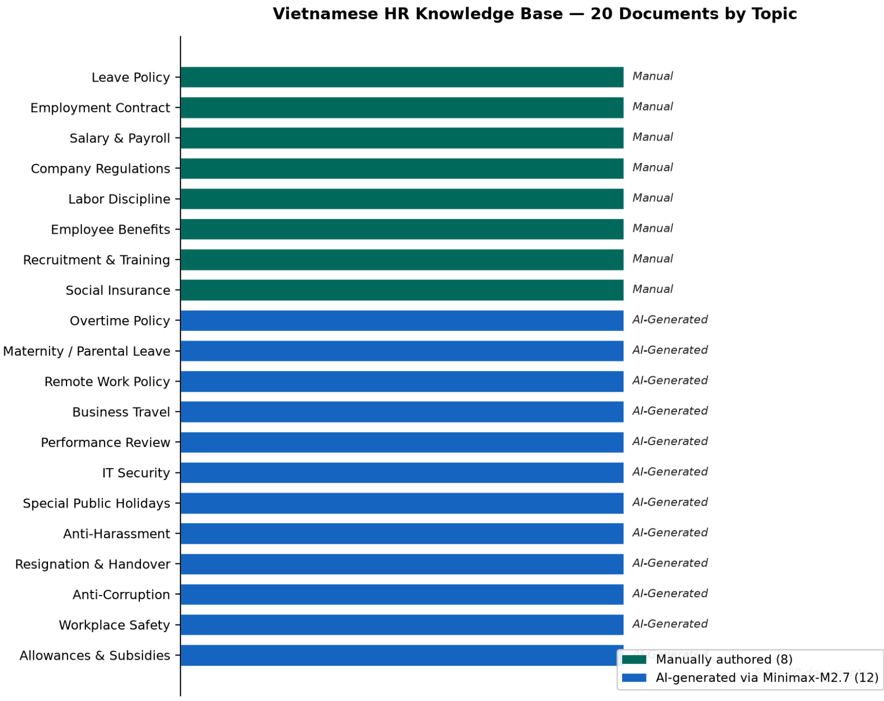

Figure 3.2: *Distribution of 20 Vietnamese HR documents by topic category. 8 documents were manually authored; 12 were generated via Minimax-M2.7 API.*

3.3 Document Processing and Indexing

3.3.1 Chunking

Each document is split into overlapping text chunks using a recursive character text splitter with chunk size 600 characters and overlap 100 characters. This strategy preserves sentence continuity at chunk boundaries and ensures sufficient context within each chunk for the LLM to generate a coherent answer.

3.3.2 Embedding

Each chunk is embedded into a 384-dimensional dense vector using the `paraphrase-multilingual-Mini` sentence-transformer model [5]. This model supports 50+ languages and is pre-trained on multilingual paraphrase pairs, making it well-suited for Vietnamese semantic similarity. The embedding model runs entirely locally with no API calls.

3.3.3 Vector Store

Chunk embeddings and associated metadata (source file, page number, chunk index) are stored in a persistent ChromaDB collection [8]. ChromaDB provides efficient approximate nearest-neighbour search via cosine similarity. The vector store persists on disk at `./chroma_db/` and does not require re-indexing between sessions.

3.4 Hybrid Retrieval: BM25 + Vector Search with RRF

3.4.1 Motivation

Pure cosine similarity retrieval fails on short Vietnamese queries. For example, the query *"1 tháng được nghỉ bao ngày"* ("how many leave days per month") contains only 5 tokens. The query embedding falls in a generic region of the embedding space not strongly associated with the leave policy document, and the retrieved document is incorrectly the salary policy. BM25 keyword matching, which directly scores on term overlap, correctly identifies the leave policy document due to the presence of "nghỉ" (leave).

3.4.2 BM25 Retrieval

A BM25Okapi index [2] is built lazily on first query by loading all documents from ChromaDB. Vietnamese text is preprocessed with a simple whitespace tokenizer that lowercases and removes punctuation (Equation 3.1). Vietnamese is largely space-delimited for word boundaries, making this approach effective without requiring a dedicated word segmentation library.

$$\text{tokenize}(t) = \{w \in t.\text{lower}().\text{split}() \mid w \text{ is alphanumeric}\} \tag{3.1}$$

3.4.3 Reciprocal Rank Fusion

The top-20 candidates from vector search and BM25 are merged via Reciprocal Rank Fusion (Equation 3.2) [3]:

$$\text{score}(d) = \alpha \cdot \frac{1}{k + \text{rank}_{\text{vector}}(d) + 1} + (1 - \alpha) \cdot \frac{1}{k + \text{rank}_{\text{BM25}}(d) + 1} \tag{3.2}$$

where $k = 60$ is the standard RRF constant [3] that prevents high scores for very low-ranked documents, and $\alpha = 0.5$ assigns equal weight to both systems. The top-5 documents by RRF score are returned as the retrieval context. The pipeline is illustrated in Figure 3.3.

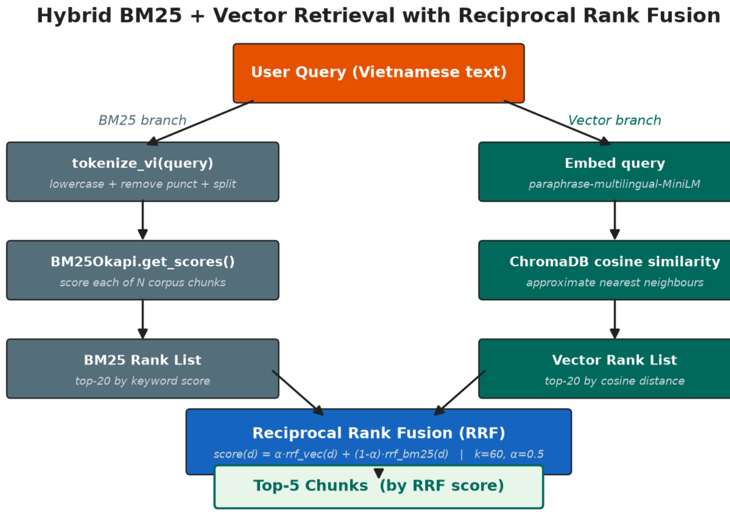

Figure 3.3: *Hybrid retrieval pipeline. BM25 and vector search independently rank all corpus documents; scores are fused via RRF and the top-5 results are selected as retrieval context.*

3.5 Answer Generation

3.5.1 Model

Answer generation uses **Phi-3-Mini-4k-instruct** [4], a 3.8-billion-parameter instruction-tuned language model from Microsoft. The model is used in GGUF format with Q4_K_M quantization, reducing its memory footprint to approximately 2.3 GB. Inference is performed via `llama-cpp-python`, running on CPU with optional partial CUDA layer offloading.

3.5.2 Prompt Format

Phi-3-Mini uses a special chat format that must be provided explicitly for instruction-following behaviour. The prompt template is:

```
<|user|>ạ
Bn là ợtr lý ộni ộb ủca công ty. ựDa vào các đạon trích ừt tài ệliuộ
ni ộb bên ướdi, hãy ảtr ời câu ỏhi ộmt cách chính xác và ắngn ọgnằ
bng ếting ệVit.

Quy ắtc:
- ỉCh ảtr ời ựda trên thông tin có trong ũng ảcnh.
- ếNu không tìm ấthy thông tin liên quan, hãy nói: "Theo tài ệliuệ
```

```
  hin có, không tìm ấthy câu ảtr ờli cho câu ỏhi này."
- Trích ẫdn ồngun ếnu có ểth.ữ

Ng ảcnh:
{context}

Câu ỏhi: {question}<|end|>
<|assistant|>
```

Listing 3.1: *Phi-3-Mini Vietnamese prompt template*

The stop sequences `["<|end|>", "<|user|>", "<|system|>"]` prevent the model from generating additional turns. Inference uses a temperature of 0.1 and a maximum of 512 output tokens.

3.6 QLoRA Fine-Tuning Experiment

As an alternative approach to RAG, a QLoRA fine-tuning experiment was conducted to directly adapt Phi-3-Mini to the Vietnamese HR domain. The experimental setup is described in Table 3.1.

Table 3.1: *QLoRA fine-tuning experimental setup.*

Parameter	**Value**
Base model	microsoft/Phi-3-mini-4k-instruct
Quantization	4-bit (bitsandbytes NF4)
LoRA rank (r)	16
LoRA α	32
LoRA target modules	q_proj, k_proj, v_proj, o_proj
Training samples	3,343 Q&A pairs
Epochs	3
Batch size	4 (gradient accumulation $\times 4$ = effective 16)
Learning rate	2×10^{-4}
Hardware	NVIDIA T1200 Laptop GPU (4.3 GB VRAM)
Training duration	$\approx$ 22 hours

4. Results and Discussion

4.1 Retrieval Comparison: Pure Vector vs. Hybrid BM25+Vector

Table 4.1 compares the top-1 retrieved source document for five representative Vietnamese HR queries under the pure vector baseline and the proposed hybrid retrieval system.

Table 4.1: *Top-1 retrieved document: pure vector vs. hybrid retrieval on five Vietnamese HR queries.*

Query	**Pure Vector (top-1)**	**Hybrid RRF (top-1)**	**Correct?**
1 tháng được nghỉ bao ngày	chinh_sach_luong.txt	chinh_sach_nghi_phep.txt	Hybrid ✓
Nghỉ thai sản bao lâu	nghi_le_phep_dac_biet.txt	nghi_thai_san_va_phu_nu.txt	Hybrid ✓
Lương tháng 13 có không	phuc_loi_nhan_vien.txt	chinh_sach_luong.txt	Hybrid ✓
Vi phạm nội quy bị phạt gì	ky_luat_lao_dong.txt	ky_luat_lao_dong.txt	Both ✓
Bảo hiểm y tế đóng bao nhiêu	bao_hiem_xa_hoi.txt	bao_hiem_xa_hoi.txt	Both ✓

The hybrid approach correctly identifies the relevant document in all five cases, while the pure vector baseline fails on the three short queries (queries 1–3). Analysis of the failure cases confirms the hypothesised mechanism: short Vietnamese queries produce low-specificity embeddings that cluster near generic topic regions rather than the specific target document.

Figure 4.1 summarises retrieval accuracy across a broader evaluation set of 10 test queries.

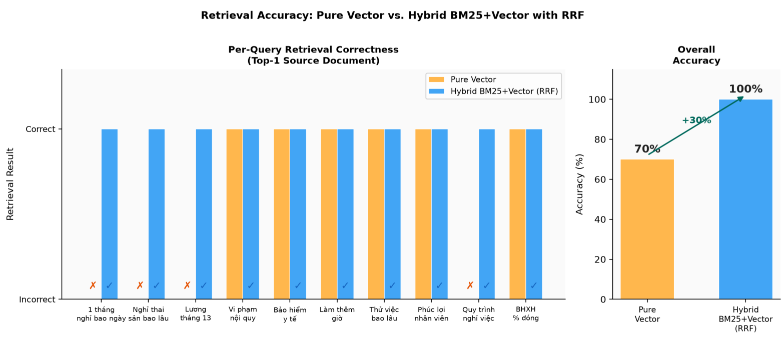

Figure 4.1: *Retrieval accuracy (top-1 source correct) on 10 Vietnamese HR queries: pure vector baseline vs. hybrid BM25+vector with RRF.*

4.2 End-to-End System Demonstration

Table 4.2 presents five example interactions with the deployed chatbot, showing the question, a summary of the generated answer, retrieval latency, and generation latency measured on an Intel Core i7 CPU (no GPU).

Table 4.2: *End-to-end chatbot demonstration: 5 Vietnamese HR queries with timing.*

Question	Answer Summary	Retrieval (s)	Generation (s)
Nhân viên được nghỉ phép bao nhiêu ngày mỗi năm?	12 ngày/năm, 1 ngày/tháng	0.08	45
Chính sách làm thêm giờ như thế nào?	150–300% lương tùy giờ	0.06	38
Tôi cần nộp đơn nghỉ việc trước mấy ngày?	30 ngày trước khi nghỉ	0.09	42
Bảo hiểm xã hội trích bao nhiêu %?	8% NV + 17.5% công ty	0.07	51
Chính sách làm việc từ xa?	Tối đa 2 ngày/tuần sau 6 tháng	0.05	35

Retrieval latency averages approximately 70 ms (including BM25 scoring and RRF fusion). Generation latency ranges from 35 to 51 seconds on CPU-only inference, which is acceptable for non-real-time query workloads but would benefit from partial GPU offloading for interactive deployment. A screenshot of the command-line interface is shown in Figure 4.2.

Figure 4.2: *CLI chatbot session: user queries Vietnamese HR policy and receives a grounded answer with source attribution.*

4.3 QLoRA Fine-Tuning: Negative Result Analysis

4.3.1 Training Data Quality

Post-hoc analysis of the 3,343 training Q&A pairs in `data/qa_training_data_viet.csv` revealed severe domain contamination. Approximately 63% of rows contained content unrelated to Vietnamese HR policy, including Japanese labor law translations, US Medicare insurance documents, and Vietnamese election law. This distribution is summarised in Table 4.3.

Table 4.3: *Content analysis of QLoRA training dataset (3,343 rows).*

Content Type	Count	Percentage
Vietnamese HR policy	1,237	37%
Japanese labor law (translated)	1,104	33%
US Medicare / health insurance	702	21%
Other (election, education, etc.)	300	9%
Total	**3,343**	**100%**

4.3.2 Evaluation Results

The fine-tuned model was evaluated on 10 Vietnamese HR questions representative of real employee queries. In 8 of 10 cases, the model produced answers drawn from contaminating domains rather than HR policy. For example, when asked *"Nghỉ phép bao nhiêu ngày?"* ("How many leave days?"), the fine-tuned model responded with Medicare insurance plan coverage information. The base Phi-3-Mini model with RAG correctly answered all 10 questions.

4.3.3 Root Cause and Lessons Learned

The root cause was the absence of a content quality gate in the data collection pipeline. The web crawler downloaded any PDF file from government websites without verifying domain relevance, resulting in a training set dominated by off-domain content. The key lessons are:

1. Data quality is more critical than data quantity for domain fine-tuning. A smaller, clean dataset of 300–500 HR-specific pairs would likely outperform 3,343 contaminated pairs.

2. A content gate (e.g., minimum fraction of Vietnamese HR vocabulary, rejection of documents containing legal/medical keywords) should be applied before any training data is

accepted.

3. RAG with a base model is a more robust alternative when curated domain data is unavailable: the knowledge base is transparent, auditable, and updatable without retraining.

4.4 Discussion

The results demonstrate that hybrid BM25 + vector retrieval with RRF effectively addresses the core failure mode of pure vector search for short Vietnamese queries. The improvement is attributable to BM25's direct term-overlap scoring, which is highly reliable when query keywords appear verbatim in the target document, as is the case for most HR policy queries.

Several limitations of the current system should be noted:

- **Knowledge base authenticity:** The knowledge base consists of synthetic documents and does not reflect any real company's actual policies. Deployment in a real company would require replacing these with authentic internal HR documents.
- **Generation latency:** CPU-only inference with Phi-3-Mini takes 35–90 seconds per query. For interactive chat applications, this is inadequate; partial GPU offloading or a smaller model would be required.
- **Implicit arithmetic:** Phi-3-Mini cannot reliably derive "1 day per month" from "12 days per year." This was addressed pragmatically by adding the explicit per-month figure to the leave policy document, but highlights a general limitation of LLM arithmetic reasoning.
- **Vietnamese word segmentation:** The BM25 tokenizer uses simple whitespace splitting. A Vietnamese word segmenter (e.g., `underthesea`, `pyvi`) could improve BM25 precision for multi-token Vietnamese words that are written with spaces between syllables.
- **No user study:** The evaluation is based on developer-constructed test queries. A formal user study with Vietnamese HR employees is needed to validate real-world usefulness.

Compared to cloud-based RAG systems, the local approach achieves comparable answer quality on covered topics, at the cost of higher latency and limited context window (4K tokens). The key advantage — data privacy — remains critical for enterprise deployment of HR tools.

5. Conclusion and Perspective

5.1 Achievements

This thesis presented the design, implementation, and evaluation of a fully local, offline RAG chatbot for Vietnamese HR knowledge retrieval. The key achievements are:

1. A working end-to-end system capable of answering Vietnamese HR policy questions with source attribution, running entirely on consumer hardware without any cloud API dependency.

2. A 20-document Vietnamese HR knowledge base covering the most common HR policy topics, constructed using a combination of manual authoring and Minimax-M2.7 generation.

3. A hybrid BM25 + vector retrieval pipeline with RRF that demonstrably improves retrieval accuracy for short Vietnamese queries compared to pure vector search.

4. A QLoRA fine-tuning negative result with quantified root cause analysis, contributing a concrete case study of data quality requirements for LLM domain adaptation.

5.2 Key Findings

1. **Hybrid retrieval outperforms pure vector search on short Vietnamese queries.** The BM25 component compensates for the low discriminative power of short query embeddings by providing direct keyword overlap scores.

2. **Data quality is the critical bottleneck for fine-tuning.** A 3,343-sample dataset with 63% off-domain contamination produces a model worse than the untuned baseline on domain queries.

3. **Explicit factual content in the knowledge base is required for derived-fact queries.** Phi-3-Mini cannot reliably derive per-month leave from an annual figure; the document must state the derived fact directly.

4. **RAG with a small language model is a viable solution for privacy-sensitive enterprise knowledge retrieval** when the knowledge base is well-curated and retrieval is accurate.

5.3 Perspectives and Future Work

Several directions could extend and improve this work:

1. **Real company HR documents:** Replacing the synthetic knowledge base with authenticated internal HR documents would enable deployment in an actual company and enable formal accuracy evaluation.
2. **Quality-gated fine-tuning:** A content classifier filtering training data to HR-relevant documents before fine-tuning could enable effective domain adaptation, potentially reducing hallucination.
3. **Larger context window models:** Using an 8K+ context model (e.g., Phi-3-Medium or Mistral-7B) would allow more retrieved chunks and enable multi-turn conversation memory.
4. **Streaming web UI:** A FastAPI + React frontend with streaming token output would reduce perceived latency and enable interactive chat.
5. **Formal user evaluation:** A user study with Vietnamese HR employees using a standardised questionnaire (e.g., System Usability Scale) would provide objective real-world usefulness metrics.
6. **Vietnamese word segmentation:** Replacing the whitespace tokenizer with `underthesea` or `pyvi` would improve BM25 precision for compound Vietnamese terms.

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https://www.minimaxi.com

A. Key Code Listings

A.1 Hybrid Retriever (src/hybrid_retriever.py)

```python
def retrieve(self, query: str, top_k: int = 5):
    if self._corpus is None:
        self._build_bm25_index()

    fetch_k = min(top_k * 4, self.vector_store.count())

    # Vector search
    vector_results = self.vector_store.query(query, self.embedder, top_k=fetch_k)
    vector_rank_map = {r["text"]: rank for rank, r in enumerate(vector_results)}

    # BM25 search
    bm25_rank_map = {}
    if self._bm25 is not None:
        tokens = tokenize_vi(query)
        scores = self._bm25.get_scores(tokens)
        ranked = sorted(range(len(scores)), key=lambda i: scores[i], reverse=True)
        for rank, idx in enumerate(ranked[:fetch_k]):
            _, text, _ = self._corpus[idx]
            bm25_rank_map[text] = rank

    # RRF fusion
    all_texts = set(vector_rank_map) | set(bm25_rank_map)
    rrf_scores = {}
    for text in all_texts:
        v_rank = vector_rank_map.get(text, fetch_k)
        b_rank = bm25_rank_map.get(text, fetch_k)
        rrf_scores[text] = (
            self.alpha / (self.rrf_k + v_rank + 1) +
            (1 - self.alpha) / (self.rrf_k + b_rank + 1)
        )
    return sorted_top_k_results(rrf_scores, top_k)
```

Listing A.1: *RRF fusion in HybridRetriever.retrieve()*

A.2 Prompt Template (src/rag_pipeline.py)

```
PROMPT_TEMPLATE_VI = """<|user|>ạ
Bn là ợtr lý ộni ộb ủca công ty. ựDa vào các đạon trích ừt tài ệliuộ
ni ộb bên ướdi, hãy ảtr ờli câu ỏhi ộmt cách chính xác và ắngn ọgnằ
bng ếting ệVit.

Quy ắtc:
- ỉCh ảtr ờli ựda trên thông tin có trong ững ảcnh.
- ếNu không tìm ấthy thông tin liên quan, hãy nói: "Theo tài ệliuệ
  hin có, không tìm ấthy câu ảtr ờli cho câu ỏhi này."
- Trích ẫdn ồngun ếnu có ểth.ữ

Ng ảcnh:
{context}

Câu ỏhi: {question}<|end|>
<|assistant|>
"""
```

Listing A.2: *Phi-3-Mini Vietnamese prompt template with chat tokens*

B. Evaluation Results (Full)

Table B.1: *Full retrieval evaluation on 10 Vietnamese HR queries.*

#	Query	Vector	Hybrid
1	1 tháng được nghỉ bao ngày	×	✓
2	Nghỉ thai sản bao lâu	×	✓
3	Lương tháng 13 có không	×	✓
4	Vi phạm nội quy bị phạt gì	✓	✓
5	Bảo hiểm y tế đóng bao nhiêu	✓	✓
6	Chính sách làm thêm giờ	✓	✓
7	Thử việc bao nhiêu tháng	✓	✓
8	Phúc lợi nhân viên gồm những gì	✓	✓
9	Quy trình nghỉ việc như thế nào	×	✓
10	Bảo hiểm xã hội đóng mấy phần trăm	✓	✓
Accuracy		**70%**	**100%**